



Sir Syed University of Engineering & Technology

Faculty of Computing and Applied Sciences

Department of Computer Science and Information Technology

Mid-Term Examinations (Spring 2023)

Course Code with Title	HS-412 Economics and Management		Program	Bachelor of Computer Science
Instructor	Syeda Uzma Shoaib		Semester	7 th
Start date & Time	May 17, 2023 at 11:30AM	End Time	1:00 PM	
Maximum Marks	30			

Attempt All Questions.

Q.1.

(05) + (05)

- a) Explain how the principles of economics can help to solve the basic economic problems that societies are facing. Use examples to support your answer.
- b) Identify the characteristics of a perfectly competitive and monopoly market.

Q.2.

(04) + (06)

- a) Predict that the relationship between Money and Time is dynamic and always changing. Show the factors that lead to change in the value of money and decrease purchasing power of the buyers.
- b) A Real Estate agent sold a building. The firm is expected to receive payment in four installments. The Buyer has paid \$10,000 at purchase time. The buyers would pay \$15,000 at the end of the first year \$18,000, at the end of the second year \$21,000, and at the end of 3rd year. Find the present worth of the amount received. The Real Estates wishes to use a Discount Rate of 7%.

Q.3.

(04) + (06)

- a) Discuss the objectives of Benefit-Cost analysis and the Methods of determining the B/C ratio in detail.
- b) A Project Director decides to provide an access to road and transportation to the project. In response, Project Director received two comparable alternates that include the following details:

<u>Break-up</u>	<u>Alternate A</u>	<u>Alternate B</u>
Initial cost	\$ 20,000	\$ 24,000
Project life	5 years	5 years
SV	\$ 4,000	\$ 4,000
Annual benefits	\$ 10,000	\$ 10,000
Annual O&M	\$ 4,400	\$ 4,400
Interest rate	8%	8%

- a) Determining the B/C ratio by Conventional Method.
- b) Do you recommend Alternate A or Alternate B.
- c) Give reasons in support.



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Psychology)
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Mid Term Examinations (Spring 2023)

Course Code with Title	CS-467: Software Project Management		Program	Bachelor of Computer Science (BSCS)
Instructor	Engr. M.Nadeem, Ms. Anam Siddiqui & Mr. M. Faheem		Semester	7 th
Start date & Time	May 19, 2022 at 11:30 AM	End Time	01:00 PM	
Maximum Marks	30			

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:
Attempt All Questions.

Q.1.

(06)

Suppose ABC Infrastructure, a manufacturing company, uses the expected monetary value (EMV) method to analyze a project's risk score and develop a contingency reserve. Calculate contingency reserve by using following table.

Risk-ID	Risk	Probability	Impact
1	Risk A (Threat)	12%	\$24,000
2	Risk B (Threat)	58%	\$15,000
3	Risk C (Threat)	9%	\$54,000
4	Risk D (Threat)	18%	\$21,000
5	Risk E (Opportunity)	23%	\$-22,000
6	Risk F (Threat)	65%	\$72,000

Q.2.

(06)

ABC software house will initially investment \$20000 for its software project with discount rate is 14%. Project represents a series of cash flows as follows

Year	Cash Inflows
1	\$ 2,300
2	\$ 3,300
3	\$ 3,200
4	\$ 6,400
5	\$ 3,500

- Compute net present value (NPV) of the project.
- Is project acceptable to company? Why



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Q.3.

(06)

You have taken over a software development project 10 months after it started, and need to find out project stakeholders and roles. What kind of document do you need and why? Discuss it?

Q.4.

(06)

PMI has developed a project management framework which provides a basic structure and knowledge areas to describe project management body of knowledge and practices in terms of their component processes. These processes have been organized into 10 knowledge areas. List down the each knowledge area with brief description?

Q.5.

(06)

Draw the project network diagram (PDM) and find the critical path by using following project plan.

Activity	Predecessor	Duration in (Days)
A	None	11
B	A	10
C	A	9
D	B	10
E	C	30
F	D	8
G	E & F	8
H	G	20
I	G	8
J	H & I	9
K	J	3



Mid Term Examinations (Spring 2023)

Course Code with Title	CS-469: IS Audit	Program	BS CS
Instructor	Farhana Ashraf & Sobia Asher	Semester	7 th
Start date & Time	May 18, 2023 at 11:30 AM	End Time	01:00 PM
Maximum Marks	30		

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.
- Understanding 'audit' terms and language is part of your skill set

Q.1.

(06)

As part of the internal audit function in a reputable bank, in what manner would you manage the peer pressure in routine audit tasks keeping your independence intact? Justify your answer with an example.

Q.2.

(06)

Your firm is engaged in the audit of an information system processing facility. You have been assigned the task of evaluating the effectiveness of the logical and environmental controls related to 'Power and fire hazards'

Required:

Specify the questions that you would ask and the matters that you would like to observe to assess the effectiveness of controls related to the above areas.

Q.3.

(3+3)

Give reasons to justify following statements. Support your answer with suitable examples.

- a) Shared responsibility prevents the IT department from independently making and later being solely responsible for poor decisions.
- b) Without removing the old access rights, there could be a potential segregation of duties issue.



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Q.4. (06)

The IS auditor is conducting an audit of the server room of a medium sized hospital. The hospital has deployed customized user access control software developed by a recently. It was noted that:

- The hospital has ninety computers, including three servers, which are connected through LAN. There is one vacant network point in each department and in each of the four wards.
- Senior doctors use these points for connecting their laptops to the network to view patients' history.
- Internet facility is available to all users through LAN.
- Entrance to server room is through IT Manager's room.

Required:

What physical access controls should be recommended by the IS auditor which could help to ensure the physical security of the server room.

Q.5. (06)

As part of an IS audit in a medium size financial institution, you are documenting the IT general controls and mapping them with the best practices. You have noted that all the users have access to the entire printing options. The client is of the view that this practice makes the system user friendly and enhances its operating efficiency. The client also believes that it would not create any threat.

Required:

Comment on the arguments provided by the client and state what action would you take.





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Mid Term Examinations (Spring 2023)

Course Code with Title	HS-410: Professional Practices	Program	BS (CS)
Instructor	Mr. S. Mohsin Ali Naqvi / Mr. Shahbaz	Semester	7th
Start date & Time	May 16, 2023 at 11:30 AM	End Time	01:00PM
Maximum Marks	30		

IMPORTANT INSTRUCTION:

Attempt All Questions.

Q.1

(03+02)

Suppose you have joined an organization as a Software Engineer. As a professional person you have the responsibilities of developing not only new software but also updating existing software. In this light:

- (a) Discuss briefly any six **traits** which you have to follow as an employee of the company.
- (b) Differentiate between **Law** and **Ethics**. Also give one **example** with each.

Q.2

(08)

Assuming that you are the owner of a software house and twenty five software developers and programmers are working under your control.

Briefly explain each IEEE code of Ethics which you will implement among your employees.

Q.3

(03+06+08)

- (a) A software is an **Intellectual Property**. What do you understand by this statement? Also give any four examples of Intellectual Property.
- (b) What do you understand by **Software Risks**? Enlist all possible such risks which a software engineer may face during his/her software development life cycle.
- (c) Explain shortly **Cybercrime** and list down different forms of Cybercrime. Discuss briefly the word **Cyber Security** and how can you protect your software and data from cybercrime?